SchoolPilot

PRODUCT FEATURE & ARCHITECTURE SPECIFICATION

Cloud-Based K-12 School Management System

Version 2.1 • October 2026 • Software-Only • Zero Hardware Required

Executive Overview: SchoolPilot is a next-generation, cloud-based school management platform engineered specifically for private K-12 day and boarding schools in Nigeria. Operating strictly under a **Software-Only** architectural core, SchoolPilot eliminates expensive biometric scanners, RFID readers, and specialized hardware. It turns standard smartphones, browsers, and existing office PCs into an integrated hub for administration, AI-assisted teaching, parent engagement, computer-based testing (CBT), and automated fee processing.

1. Zero-Hardware & Software-Only Architecture

Traditional school management platforms require costly biometric readers, facial scanners, and smart cards. SchoolPilot completely replaces hardware procurement with standard mobile and desktop web capabilities:

- **QR Code Camera Attendance:** Teachers and gatekeepers scan student badges or mobile passes directly using the built-in smartphone camera view for real-time attendance logging.
- **GPS Geofenced Staff Clock-In:** Staff log arrival and departure via smartphone GPS verification, enforcing authentic clock-ins strictly within school premises.
- **Real-Time Bus Location Broadcasting:** Bus drivers run the mobile app during active routes, allowing parents to track bus arrival live without vehicle GPS tracker hardware.
- **Phone Camera Library Barcoding:** Library check-in and checkout uses the phone camera to scan standard book barcodes, removing the need for handheld barcode guns.
- **Printable Student & Staff ID Cards:** Generates high-resolution, print-ready PDF ID cards complete with QR codes and school branding, printable on standard office printers.
- **Offline CBT Lab Client (pilot):** One school laptop serves the sealed paper to existing lab PCs over the lab's own network, with no internet during the exam; answers upload afterwards.

2. Core Academic & Administrative Management (K-12 Focused)

Designed specifically around Nigerian K-12 academic structures (Sessions, Terms, Classes, Arms/Streams) rather than generic higher-education faculty concepts.

- **Nigerian Academic Structure Native:** Full support for 3-Term sessions, Continuous Assessment (CA) weightings, Termly Broadsheets, JSS/SS divisions, and national curriculum standards (WAEC, NECO, JAMB).
- **Student Information System (SIS):** Centralized digital student dossiers tracking profiles, guardian linkages, medical notes, academic history, and NDPA 2023 compliant data protection.
- **Automated Conflict-Free Timetabling:** Intelligent schedule generator balancing subject periods, teacher availability, and classroom allocations without scheduling overlaps.
- **Unified Multi-Tenant Web & Mobile App:** Next.js web portal and one Flutter mobile app with role-based access for Admins, Bursars, Teachers, Students and Parents, each school on its own subdomain.
- **Boarding & Campus Operations:** Hostel rooms and beds, exeats with parent consent, roll calls, incident reports, hostel fees on the term invoice, sick bay, library, transport and ID cards.

3. AI-Powered Teaching & Learning Ecosystem

- **AI Teacher Copilot:** Drafts lesson plans, exam questions, homework ideas, translations and performance summaries on the Nigerian curriculum, for the teacher to edit.
- **Choice of AI Provider:** Claude, ChatGPT or Gemini, chosen by the school with a quality level; a daily spending cap per school, and a child's name is never sent to the provider.
- **AI Report Card Remarks with Human Oversight:** Drafts constructive report card remarks from subject and score. AI comments stay in *pending_approval* until a teacher approves or edits them.

• **AI Study Tutor:** Guides students through problems rather than handing over answers, within the sections, subjects and daily limits the school sets, and locked during CBT exams.

• **Dashboards & Alerts:** Principal, teacher, student and parent dashboards: attendance, fees collected, results awaiting approval and students whose scores are slipping.

4. Computer-Based Testing (CBT) & Assessment Engine

• **Rich Media & Maths Questions:** Objective, theory and comprehension-passage questions with images and maths rendered offline; theory answers go to the teacher to mark.

• **Exam Hall Controls:** Per-candidate question and option shuffling, timed sittings with autosave, session events logged for review, and automatic objective marking.

• **WAEC, NECO & BECE Practice:** Schools set exam-style practice papers from their own question bank, tagged by subject, topic, difficulty and exam body, for SS3 and JSS3 candidates.

5. Fee Management & Financial Integration

• **Integrated Payment Gateways:** Native integration with Paystack and Flutterwave enabling tuition payment via Debit Card, Bank Transfer, or USSD with automated instant receipts.

• **Parent Financial Ledger & Invoicing:** Real-time itemized breakdown of tuition, uniforms, bus services, and extracurricular fees with partial payment support and clear balance tracking.

• **School Store (Uniforms, Books & Supplies):** Optional stock of what the school hands to students, by size and edition; new-student packs issued at registration; charged items added to that term's invoice; parents confirm receipt in the app, and anything missing comes back to the school.

• **WhatsApp & SMS Communication Hub:** Automated payment reminders, attendance alerts, exam results, and school announcements sent directly to parents' mobile phones.

6. User Roles & System Permissions Matrix

Role	Primary Responsibilities & Access Scope	Key Features Available
Super Admin	Platform operators & system maintenance	Multi-school onboarding, per-term billing, AI provider and spend, result PINs, audit and compliance
School Admin	Principal, Proprietor, & Front Office	Staff/Student records, fee structures, timetables, broadsheet approval, AI insight dashboard
Bursar	Finance office	Term invoices, collecting payments with receipts, scholarships, hostel charges, school store, accounting
Teacher	Academic staff & Subject Instructors	Attendance scanning, CA grading, AI lesson planner, AI report card remark approval
Student	Enrolled learners (Primary - SS3)	Printable timetable, homework, CBT, released results, progress, AI study tutor, profile
Parent	Guardians & Sponsors	Fees and payments, attendance, homework, report cards, hostel, confirming items from the school store, pickup authorisation and consent

READY TO TRANSFORM YOUR SCHOOL MANAGEMENT?

Zero hardware • From NGN 300 per student per term (30-day free trial) • NGN 500 with Result Management • Boarders NGN 1,000